lab #8

Problema 1

Parte a)

$i = \lfloor n/2 \rfloor$, $i = n$

$N_i = n - \lfloor \frac{n}{2} \rfloor + 1 = \lceil \frac{n}{2} \rceil + 1$

$j + \lfloor \frac{n}{2} \rfloor \leq n \rightarrow j \leq \lceil \frac{n}{2} \rceil$

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$j = 1$

$N_j = \lceil \frac{n}{2} \rceil$

$k \rightarrow 1, 2, 4, 8, \ldots, 2^r$

$2^r \leq n \longrightarrow r \leq \log_2 n$

$N_k = \lfloor \log_2 n \rfloor + 1$

$C(n) = (\lceil \frac{n}{2} \rceil + 1)\lceil \frac{n}{2} \rceil (\lfloor \log_2 n \rfloor + 1)$

$T(n) = \Theta(n) * \Theta(n) * \Theta(\log n) = \Theta(n^2 \log n)$

$T(n) = O(n^2 \log n)$

Problema 2

Parte a)

$n \leq 1 = T(n) = \Theta(1)$

$n > 1 = N_i = n$

$N_j = 1$

$C(n) = \sum_{i=1}^{n} 1 = n$

$T(n) = O(n)$

Problema 3

Parte a)

$i = 1, 2, \ldots, \lfloor \frac{n}{3} \rfloor$

$N_i = \lfloor \frac{n}{3} \rfloor$

$j = 1, 5, 9, 13, \ldots$

$j = 1 + 4r$

$1 + 4r \leq n \rightarrow r \leq \frac{n-1}{4}$

$N_j = \lfloor \frac{n-1}{4} \rfloor + 1 = \lceil \frac{n}{4} \rceil$

$C(n) = \lfloor \frac{n}{3} \rfloor \lceil \frac{n}{4} \rceil$

$T(n) = \Theta(n) * \Theta(n) = \Theta(n^2)$

$T(n) = \Theta(n^2)$

Problema 4

a) $C_{mejor}(n) = 1$

$T_{mejor}(n) = \Theta(1)$ } mejor caso

$C_{Promedio}(n) = \frac{1 + 2 + \cdots + n}{n}$

$C_{promedio}(n) = \frac{n(n+1)}{2n} = \frac{n+1}{2}$

$T_{promedio}(n) = \Theta(n)$ } caso Promedio

$C_{Peor}(n) = n$

$T_{Peor}(n) = \Theta(n)$ } Peor caso

b) $C_{mejor}(n) = 1$

$T_{mejor}(n) = \Theta 1$

$C_{promedio}(n) = \frac{\sum_{d=1}^{h} d\, 2^{d-1}}{2^h - 1}$

$$S = 1 + 2*2 + 3*2^2 + \dots + h\,2^{h-1}$$

$$S = h\,2^h - (1 + 2 + \dots + 2^{h-1})$$

$$S = h\,2^h - (2^h - 1) = (h-1)\,2^h + 1$$

$$C_{Promedio}(n) = \frac{(h-1)2^h + 1}{2^h - 1} = h - 1 + \frac{h}{2^h - 1}$$

$$h = \log_2(n+1)$$

$$T_{Prom}(n) = \Theta(\log n)$$

$$\frac{n}{2^r} \leq 1$$

$$n \leq 2^r \rightarrow r \geq \log_2 n$$

$$n \geq 1$$

$$C_{peor}(n) = \lfloor \log_2 n \rfloor + 1$$

$$T_{peor}(n) = \Theta(\log n)$$

C) $$C(n) = 2C\left(\frac{n-1}{2}\right) + (n-1)$$

$$n = 2^h - 1$$

$$h = \log_2(n+1)$$

$$\frac{n+1}{2^d} - 1$$

$$2^d\left(\frac{n+1}{2^d} - 2\right) = (n+1) - 2^{d+1}$$

$$C_{mejor}(n) = \sum_{d=0}^{h-1} \left((n+1) - 2^{d+1}\right)$$

$$C_{mejor}(n) = h(n+1) - 2(2^h - 1)$$

$$C_{mejor}(n) = (n+1)\log_2(n+1) - 2n$$

$$T_{mejor}(n) = \Theta(n \log n)$$

Caso Promedio

$$C_n = (n-1) + \frac{1}{n}\sum_{k=0}^{n-1} C_k$$

$$nC_n = n(n-1) + \sum_{k=0}^{n-1} C_k$$

$$(n-1)C_{n-1} = (n-1)(n-2) + \sum_{k=0}^{n-2} C_k.$$

$$nC_n - (n-1)C_{n-1} = 2(n-1) + 2C_{n-1}$$

$$nC_n = (n+1)C_{n-1} + 2(n-1)$$

$$\frac{C_n}{n+1} = \frac{C_{n-1}}{n} + \frac{2(n-1)}{n(n+1)}$$

$$\frac{C_n}{n+1} = \sum_{k=1}^{n} \frac{2(k-1)}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{4}{k+1} - \frac{2}{k}\right)$$

$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

$$\frac{C_n}{n+1} = 4(H_{n+1} - 1) - 2H_n = 2H_n + \frac{4}{n+1} - 4$$

$$C_n = 2(n+1)H_n - 4n$$

$$H_n = \Theta(\log n)$$

$$T_{promedio}(n) = \Theta(n \log n)$$

0 y $n-1$ = Peor caso

$$C_{peor}(n) = C_{peor}(n-1) + (n+1)$$

$$C_{peor}(n) = (n-1) + (n-2) + \cdots + 1$$

$$C_{peor}(n) = \frac{n(n-1)}{2}$$

$$T_{peor}(n) = \Theta(n^2)$$

5) a) si $f(n) = \Theta(g(n))$ y $g(n) = \Theta(h(n))$ entonces $h(n) = \Theta(f(n))$

$c_1 g(n) \leq f(n) \leq c_2 g(n)$

$d_1 h(n) \leq g(n) \leq d_2 h(n)$

$c_1 d_1 h(n) \leq f(n) \leq d_2 h(n)$

$\frac{1}{c_2 d_2} f(n) \leq h(n) \leq \frac{1}{c_1 d_1} f(n)$

$h(n) = \Theta(f(n))$

Verdadero

b) si $f(n) = O(g(n))$ y $g(n) = O(h(n))$ entonces $h(n) = \Omega(f(n))$

$0 \leq f(n) \leq c_1 g(n)$

$0 \leq g(n) \leq c_2 h(n)$

$f(n) \leq c_1 c_2 h(n)$

$h(n) \geq \frac{1}{c_1 c_2} f(n)$

$h(n) = \Omega(f(n))$

Verdadero

$T_{tupla}(n) = \Theta(n)$

$N(n) = \sum_{i=0}^{n-1} (n - i - 1) = \frac{n(n-1)}{2}$

$l = j - i$

$T_{i,j} = \Theta(j - i)$

$f(n) = \Theta(n) + \sum_{i=0}^{n-1} \sum_{j=i+1}^{n-1} \Theta(j - i)$

$\sum_{i=0}^{n-1} \sum_{j=i+1}^{n-1} (j - i) = \sum_{l=1}^{n-1} l(n - l)$

$= n \sum_{l=1}^{n-1} l - \sum_{l=1}^{n-1} l^2$

$$= n\left(\frac{n(n-1)}{2}\right) - \frac{(n-1)\,n(2n-1)}{6}$$

$$\frac{n(n-1)(3n-(2n-1))}{6} = \frac{n(n-1)(n+1)}{6}$$

$$\frac{n^3-n}{6}$$

$f(n) = \Theta(n) + \Theta(n^3) = \Theta(n^3)$

False $f(n) = \Theta(n^3)$